

Source: Virtual_Project_Management_Consultant.ipynb

Source notebook exported for Simplilearn submission. File: Virtual_Project_Management_Consultant.ipynb

Cell 1 [MARKDOWN]

Virtual Project Management Consultant

Course-end Project 2 | IITK AIML Core: Essentials of Generative AI, Prompt Engineering & ChatGPT

Objective: Design and refine prompts so ChatGPT acts as a virtual project management consultant ? covering

Cell 2 [MARKDOWN]

1. Domain Selection & Research

Domain: Software / digital product project management (Agile + hybrid delivery)

Common PM tasks researched:

- Project planning (charter, scope, WBS/roadmap, milestones)
- Risk management (identify, score, mitigate, escalate)
- Team collaboration (RACI, ceremonies, conflict, remote rituals)
- Performance tracking (KPIs, burndown, OKRs, status reporting)

Methodologies referenced: Scrum, Kanban, PMI risk register practices, RACI, RAID logs

Tools context ChatGPT should know: Jira/Linear-style boards, Confluence/Notion docs, Slack updates

Cell 3 [MARKDOWN]

2. Interaction Scenarios

Cell 4 [CODE]

```
scenarios = [  
    {  
        "id": "S1",  
        "name": "Create a project plan",  
        "example_query": "Help me plan a 12-week mobile app MVP launch.",  
        "expected": "Phased plan, milestones, roles, assumptions, next actions",  
    },  
    {  
        "id": "S2",  
        "name": "Identify and manage risks",  
        "example_query": "What risks should I track for a cloud migration?",  
        "expected": "Risk register with probability/impact, owners, mitigations",  
    },  
    {  
        "id": "S3",  
        "name": "Enhance team collaboration",  
        "example_query": "My remote squad misses handoffs between design and eng.",  
        "expected": "Concrete rituals, RACI tweaks, communication norms",  
    },  
    {  
        "id": "S4",  
        "name": "Track project performance",  
        "example_query": "Design a weekly status pack for execs.",  
        "expected": "KPI set, red/amber/green rules, template outline",  
    },  
]  
  
import pandas as pd  
display(pd.DataFrame(scenarios))
```

Cell 5 [MARKDOWN]

3. Master Role Prompt (Virtual PM Consultant Persona)

Cell 6 [CODE]

```
PM_SYSTEM_PROMPT = """
You are "Aria", a senior virtual project management consultant.

BEHAVIOR:
- Ask up to 3 clarifying questions ONLY if critical info is missing; otherwise give a useful first-pass answer
- Structure answers with clear headings and bullet points.
- Prefer actionable steps the user can do this week.
- Call out assumptions explicitly.
- When relevant, offer frameworks (RACI, risk matrix, RAID, OKRs) without jargon overload.
- End with: (1) Top 3 next actions, (2) One risk to watch, (3) A follow-up question.

CONSTRAINTS:
- Do not invent fake company data.
- If the user domain is unclear, default to software product delivery.
- Keep responses under 350 words unless the user asks for a full template.
"""

print(PM_SYSTEM_PROMPT)
```

Cell 7 [MARKDOWN]

4. Initial vs Refined Prompts

Cell 8 [CODE]

```
prompt_iterations = [
    {
        "scenario": "S1 Planning",
        "initial": "How do I create an effective project plan?",
        "issues": "Too vague; generic textbook answer; no context",
        "refined": (
            "Act as my PM consultant. Context: 8-person cross-functional team, "
            "12-week mobile MVP, fixed launch date, budget risk medium. "
            "Produce: (a) milestone roadmap by week, (b) RACI for PM/Eng/Design/QA, "
            "(c) assumptions list, (d) top 5 risks. Keep under 300 words."
        ),
    },
    {
        "scenario": "S2 Risk",
        "initial": "What are project risks?",
        "issues": "Laundry list; not prioritized; no owners",
        "refined": (
            "Build a risk register for a SaaS cloud migration (on-prem -> AWS). "
            "Table with columns: Risk, Category, P (1-5), I (1-5), Score, Owner role, "
            "Mitigation, Trigger. Include at least 8 risks. Then highlight the top 3 by score "
            "and suggest a weekly risk review agenda (5 bullets)."
        ),
    },
    {
        "scenario": "S3 Collaboration",
        "initial": "How can my team collaborate better?",
        "issues": "Soft advice; not situation-specific",
        "refined": (
            "Remote product squad: Design finishes late, Eng starts without specs, "
            "QA finds gaps in sprint reviews. Propose a 2-week collaboration reset: "
            "ceremony changes, definition of ready/done, Slack norms, and a RACI for "
            "handoffs. Give a day-by-day checklist for week 1."
        ),
    },
]
```

```
{
  "scenario": "S4 Performance",
  "initial": "How do I track performance?",
  "issues": "Metric overload; no exec framing",
  "refined": (
    "Design a one-page weekly exec status for a platform rewrite. "
    "Include: RAG overall health, 5 KPIs (with targets), blockers, decisions needed, "
    "and a burndown narrative (3 sentences). Provide a copy-paste Markdown template."
  ),
},
]
```

```
import pandas as pd
pd.set_option("display.max_colwidth", 120)
display(pd.DataFrame(prompt_iterations[["scenario", "initial", "issues", "refined"]]))
```

Cell 9 [MARKDOWN]

5. Testing & Sample Interactions (Documented)

S1 ? Project Plan (refined prompt ? response summary)

****User:**** refined S1 prompt (mobile MVP, 12 weeks)

****Consultant response highlights:****

- Weeks 1-2 discovery & scope freeze; 3-6 build core flows; 7-9 hardening; 10-11 beta; 12 launch
- RACI: PM accountable for scope; Eng for delivery; Design for UX sign-off; QA for release gate
- Assumptions: one iOS+Android codebase, no new payment vendor
- Top risks: scope creep, App Store delay, key engineer PTO

S2 ? Risk Register excerpt

Risk	P	I	Score	Mitigation
Data migration corruption	3	5	15	Dual-run + checksum validation
Skills gap on AWS networking	4	4	16	Training + external SME
Cutover downtime overrun	3	5	15	Rehearsed runbook + rollback

S3 ? Collaboration reset (week-1 checklist sample)

1. Publish Definition of Ready (DoR) with design attach rules
2. Move handoff to a fixed Tue "spec freeze" slot
3. Add Eng shadow in design critiques
4. QA joins mid-sprint bug bash Friday
5. Single Slack channel `#mvp-handoff` for blockers only

S4 ? Exec template (Markdown)

```
```markdown
Weekly Status ? <Project> ? <Date>
RAG: Green / Amber / Red
KPIs
| KPI | Target | Actual | Trend |
Blockers
Decisions needed from execs
Narrative (3 sentences)
```
```

Cell 10 [MARKDOWN]

6. Evaluation Criteria

Cell 11 [CODE]

```
criteria = [
    {"criterion": "Accuracy", "weight": "25%", "question": "Advice matches PM best practice and stated context"},
    {"criterion": "Relevance", "weight": "25%", "question": "Answer tailored to scenario, not generic filler?"},
    {"criterion": "Clarity", "weight": "20%", "question": "Scannable structure; actionable wording?"},
    {"criterion": "Actionability", "weight": "20%", "question": "User can execute next steps this week?"},
    {"criterion": "Engagement", "weight": "10%", "question": "Ends with useful follow-up; invites dialogue?"},
]

import pandas as pd
display(pd.DataFrame(criteria))

# Self-score after refinement (1-5)
scores = {
    "S1 Planning": {"Accuracy": 5, "Relevance": 5, "Clarity": 5, "Actionability": 5, "Engagement": 4},
    "S2 Risk": {"Accuracy": 5, "Relevance": 5, "Clarity": 4, "Actionability": 5, "Engagement": 4},
    "S3 Collaboration": {"Accuracy": 4, "Relevance": 5, "Clarity": 5, "Actionability": 5, "Engagement": 5},
    "S4 Performance": {"Accuracy": 5, "Relevance": 5, "Clarity": 5, "Actionability": 5, "Engagement": 4},
}
score_df = pd.DataFrame(scores).T
score_df["Total/25"] = score_df.sum(axis=1)
display(score_df)
```

Cell 12 [MARKDOWN]

7. Iterative Optimization Log

| Iteration | Change | Result |
|-----------|--|-------------------------------|
| 1 | Vague one-liners | Generic answers |
| 2 | Added role + context (team size, timeline) | More specific plans |
| 3 | Forced output structure (tables, RACI, checklists) | Higher clarity/actionability |
| 4 | Word limit + "Top 3 next actions" footer | Better engagement, less fluff |
| 5 | Persona "Aria" + assumption callouts | Consistent consultant voice |

Prompt patterns that worked:

1. **Role** + **context** + **output format** + **constraints**
2. Ask for **examples grounded in the user's domain**
3. Require **assumptions** and **next actions** every time

Cell 13 [MARKDOWN]

8. User Feedback (Sample Group Notes)

Simulated peer review (3 classmates / colleagues):

| Feedback | Action taken |
|-----------------------------|---|
| "Risk answers need owners" | Added Owner role column in S2 prompt |
| "Too long for busy PMs" | Hard 300-350 word cap |
| "Want copy-paste templates" | S4 asks for Markdown template |
| "Follow-ups feel random" | Fixed closing trio: actions / risk / question |

Cell 14 [MARKDOWN]

9. Final Optimized Prompt Pack (Ready to Paste)

Copy these into ChatGPT after the Aria system prompt:

Cell 15 [CODE]

```
FINAL_PROMPTS = {
    "planning": (
        "Context: <team size>, <duration>, <product type>, <constraints>. "
        "Deliver a milestone roadmap, RACI, assumptions, and top 5 risks. "
```

```
        "End with Top 3 next actions, one watch-out risk, and one follow-up question."
    ),
    "risk": (
        "Build a scored risk register (P/I 1-5) for <initiative> with owners and mitigations. "
        "Highlight top 3 risks and a 15-minute weekly risk-review agenda."
    ),
    "collaboration": (
        "Diagnose this teamwork issue: <symptoms>. Propose a 2-week reset with ceremonies, "
        "DoR/DoD, RACI handoffs, and a week-1 daily checklist."
    ),
    "performance": (
        "Create a one-page weekly exec status template for <project> with RAG, 5 KPIs, "
        "blockers, decisions needed, and a 3-sentence narrative. Output in Markdown."
    ),
}

for k, v in FINAL_PROMPTS.items():
    print(f"\n=== {k.upper()} ===\n{v}")
```

Cell 16 [MARKDOWN]

10. Future Improvements

- Add industry packs (construction, healthcare IT, marketing campaigns)
- Multi-turn "project war-room" mode with persistent RAID log state
- Rubric auto-score via a second LLM judge prompt
- Integrate with real Jira/CSV exports for status generation
- Multilingual prompt variants for distributed teams

11. Project Summary

- Designed a **virtual PM consultant** persona with clear behavioral rules
- Built **4 scenario prompt families** (plan, risk, collab, performance)
- Iterated from vague questions ? context-rich, format-constrained prompts
- Evaluated with a weighted rubric; documented feedback-driven refinements
- Delivered a reusable **final prompt pack** for live ChatGPT demos

Deliverable: This notebook documenting the prompt engineering process and optimized prompts.